MILITARY SPECIFICATION

ADAPTERS, PALLET, WOODEN, FOR METAL CONTAINER FOR 105MM AND 120MM TANK AMMUNITION

This Specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the fabrication requirements, quality assurance provisions and packaging of the special adapters for tank ammunition pallets. The special adapter together with a standard wooden base will unitize the 105mm and 120mm tank ammunition cartridge.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-490	-	Cleaning and Preparation of Ferrous and Coated Surfaces for Organic Protective Coatings.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8140
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MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-171	-	Finishing of Metal and Wood Surfaces.
MIL-STD-1261	-	Welding Procedures for Constructional Steel.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation

DRAWINGS

U.S. Army Defense Ammunition Center and School

AC200000461	-	Adapter - Wooden Pallet, PA116 Container.
AC200000503	-	Adapter - Top Lifting for PA104 Containers Ammunition Pallet.

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-government documents which is current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-E8	-	Tension Testing of Metallic Materials, Standard Method of
ASTM-D3951	-	Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials. 1926 Race Street, Philadelphia, PA 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Adapters. The adapter assemblies shall comply with all requirements specified on drawings (Dwgs) AC200000461 and AC200000503 and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Welding. The welding shall comply with the requirements of MIL-STD-1261.

3.3.1 Tensile strength. The tensile strength of the weld shall equal or exceed the tensile strength of the material.

3.4 Painted surface. The painted surfaces of the adapters shall comply with the requirements of TT-C-490 and with instruction in MIL-STD-171.

3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.6 Workmanship.

3.6.1 Adapters. The adapters shall be regular, smooth, and free from wrinkles, cracks, rough spots, burrs, sharp edges and any other defect that might affect the serviceability, durability, safety and appearance of the adapters.

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3.6.2 Paint coating. The finish coat shall cover all required surfaces and be continuous with the possible exception of a few light scratches not exposing the base metal or the primer coat.

3.6.3 Parts. All parts shall be free of chips, dirt, grease, rust and foreign material. The cleaning method used shall not be injurious to any of the parts nor shall any of the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in this specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of 6 assemblies as per Dwg. AC200000503 and 6 assemblies as per Dwg. AC200000461, as applicable.

4.3.2 Inspection to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE: Adapters		SHEET 1 OF 1		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Adapter for PA116 Container (Dwg. AC200000461)				
	a. Examination for Defects	6		3.2	4.4.2.1
	Adapter for PA117 Container (Dwg. AC200000503)				
	a. Examination for Defects	6		3.2	4.4.2.1

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-70753 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Adapter-Wooden Pallet, PA116 Container or Adapter Top Lifting for PA104 Containers Ammunition Pallet	1 OF 1	AC200000461/AC200000503
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Material compliance	(a)			Certification
102	Overall width		1.0%	3.2	Gage
103	Weld tensile			3.3.1	4.5.1
104	Marking, missing, incorrect or illegible		1.0%	3.2	Visual
Minor					
201	Protective finish damaged		1.0%	3.2	Visual
202	Paint thickness		1.0%	3.2	Gage
203	Poor workmanship		1.0%	3.6	Visual

NOTES:
(a) With each lot of material submitted.

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4.4.3 Testing.

4.4.3.1 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Tensile strength. The tensile strength shall be determined in accordance with ASTM-E8. One adapter shall be randomly selected from the lot. Three specimens of the lowest gage material thickness shall be selected from the adapter for determination of the material tensile strength. Also, three specimens shall be randomly selected at right angles to the weld and the tensile strength of the weld shall be determined. Each of these values shall be compared with the lowest determined value of the material tensile strength for compliance with requirement of 3.3.1. If any of the samples fails to meet the requirement, the lot shall be rejected.

5. PACKAGING

5.1 Preservation and packaging. None required.

5.2 Packing.

5.2.1 Commercial. The adapter shall be packed in accordance with ASTM-D-3951.

5.3 Marking.

5.3.1 Commercial. Shipping marking shall comply with ASTM-D3951-82 and the following additional instructions. No shipping marking shall be placed directly on the adapters. Commercial shipping tags attached with wire shall be used unless otherwise specified. The following information shall be shown: Item name and model number, part number and revision, contract number, and date, name and address of manufacturer. If adapters are bundled, boxed or bulk-contained, the gross weight and quantity shall be included. The above information may be placed on bulk-containers or boxes by labeling or stenciling.

5.4 Shipping. When adapters from more than one lot are shipped at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing lots in transit.

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6. NOTES

6.1 Intended use. These adapters are intended for use with the pallet base and with the metal tank ammunition container for the 105mm and 120mm tank ammunition cartridge.

6.2 Ordering data. see MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Subject term (key word) listing.

Ammunition
Cannon
Cartridge
Container
Metal
Pallet
Tank
Adapters

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 8140-A766)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-70753	ADAPTERS, PALLET, WOODEN, FOR METAL CONTAINERS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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